

LAB EXPERIMENT RECORD

Set 13: Geographic Data Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze geographic city data using a map chart, scatter plot, and table in R, and to build an interactive R Shiny dashboard combining all three.

Dataset: Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Note: The dataset does not give the location of the cities. To place them on a map, the programs use assumed coordinates for illustration: City A (39.7° N, 105.0° W), City B (41.9° N, 87.6° W), and City C (33.4° N, 112.1° W). Temperature is taken in °F and elevation in feet.

PROGRAM 1: Map Chart of City Distribution

Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

```
# install.packages("maps") # run once if maps is not installed
library(maps)

# Create dataset for Set 13
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)

print(geo_data)

# Assumed coordinates (the dataset does not give city locations)
geo_data$Latitude <- c(39.7, 41.9, 33.4)
```

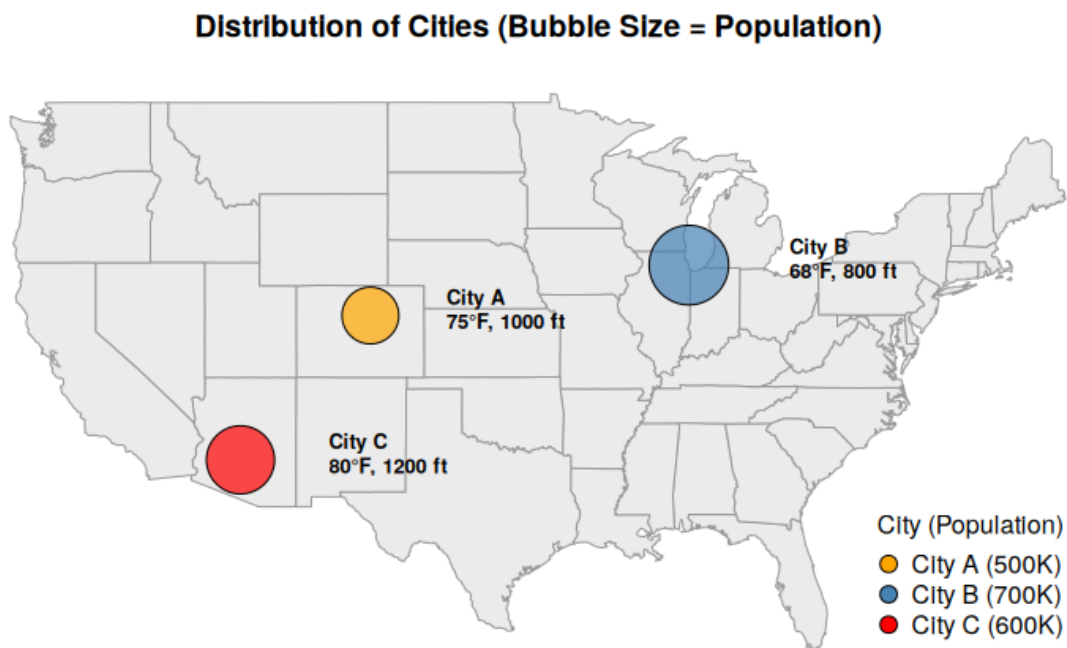
```

geo_data$Longitude <- c(-105.0, -87.6, -112.1)

# 1. Map Chart of City Distribution
map("state", fill = TRUE, col = "gray92", border = "gray60")
title(main = "Distribution of Cities (Bubble Size = Population)")

temp_cols <- c("orange", "steelblue", "red") # by temperature: 75, 68, 80
points(geo_data$Longitude, geo_data$Latitude,
       pch = 21,
       bg = adjustcolor(temp_cols, alpha.f = 0.7),
       col = "black",
       cex = geo_data$Population / 100000)
text(geo_data$Longitude + geo_data$Population / 150000, geo_data$Latitude,
     labels = paste0(geo_data$City, "\n", geo_data$Avg_Temperature, "\u00B0F, ",
                     geo_data$Elevation, " ft"),
     pos = 4, cex = 0.8, font = 2)
legend("bottomright",
      legend = paste0(geo_data$City, " (", geo_data$Population / 1000, "K)"),
      pt.bg = temp_cols, pch = 21, pt.cex = 1.5, bty = "n",
      title = "City (Population)")

```



Output 1: Distribution of Cities on the Map

Observation: The three cities are shown at their assumed locations, with bubble size representing population. City B has the largest bubble (700,000 people) and City A the smallest (500,000). Each city is labelled with its average temperature and elevation, so City C stands out as both the warmest (80°F) and the highest (1,200 ft).

PROGRAM 2: Scatter Plot of Temperature vs Population

Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

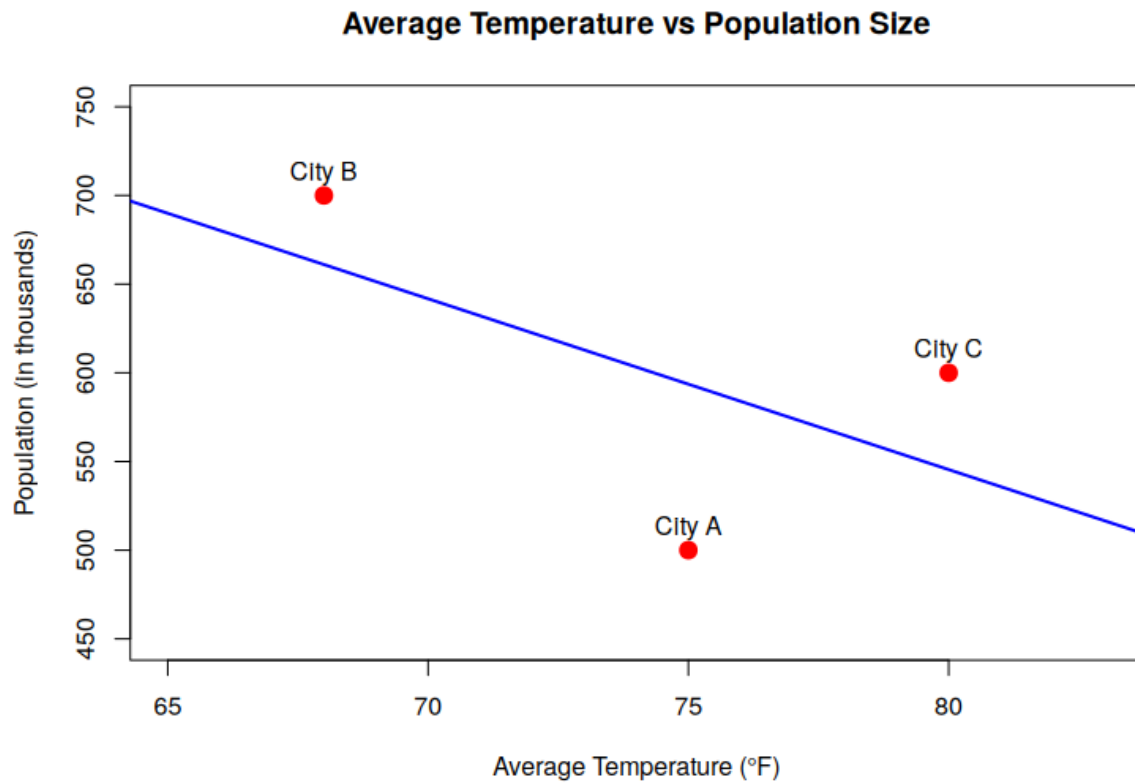
```
# Create dataset for Set 13
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)

print(geo_data)

# 2. Scatter Plot of Average Temperature vs Population
plot(geo_data$Avg_Temperature, geo_data$Population / 1000,
     col = "red",
     pch = 19,
     cex = 1.5,
     xlim = c(65, 83),
     ylim = c(450, 750),
     xlab = "Average Temperature (\u00B0F)",
     ylab = "Population (in thousands)",
     main = "Average Temperature vs Population Size")
text(geo_data$Avg_Temperature, geo_data$Population / 1000,
     labels = geo_data$City, pos = 3)

# Add a trend line
abline(lm(Population / 1000 ~ Avg_Temperature, data = geo_data), col = "blue", lwd = 2)

# Correlation between temperature and population
cor(geo_data$Avg_Temperature, geo_data$Population)
```



Output 2: Average Temperature vs Population Size

Insight: The trend line slopes downward and the correlation coefficient is about -0.58 , a moderate negative relationship. The coolest city (City B, 68°F) has the largest population, while the warmer cities have fewer people. However, City C (80°F) has more people than City A (75°F), so the pattern is not consistent. With only three cities, temperature cannot be said to decide population size.

PROGRAM 3: Table of Geographic Data

Build a table to display the geographic data for each city. Label the table elements.

```
# Create dataset for Set 13
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)

print(geo_data)

# 3. Table of geographic data
geo_table <- geo_data
geo_table$Population <- formatC(geo_data$Population, format = "d", big.mark = ",")
colnames(geo_table) <- c("City", "Population", "Avg. Temperature (F)", "Elevation (ft)")

cat("=== GEOGRAPHIC DATA TABLE ===\n")
print(geo_table, row.names = FALSE)
```

```
=== GEOGRAPHIC DATA TABLE ===
> print(geo_table, row.names = FALSE)
  City Population Avg. Temperature (F) Elevation (ft)
City A      500,000             75         1000
City B      700,000             68          800
City C      600,000             80         1200
>
```

Output 3: Geographic Data Table

Observation: The table shows the population, average temperature, and elevation of each city with clearly labelled columns and units. City B has the highest population and the lowest temperature and elevation.

PROGRAM 4: Interactive Dashboard for Geographic Data

Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)
library(maps)

# Dataset for Set 13 (with assumed coordinates)
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200),
  Latitude = c(39.7, 41.9, 33.4),
  Longitude = c(-105.0, -87.6, -112.1)
)
city_cols <- c("City A" = "orange", "City B" = "steelblue", "City C" = "red")

ui <- fluidPage(
  titlePanel("Geographic Data Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("city", "City:", choices = geo_data$City, selected = geo_data$City)
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("mapPlot", height = "300px")),
        column(6, plotOutput("scatterPlot", height = "300px"))
      ),
      h4("Geographic Data Table"),
      tableOutput("geoTable")
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({ subset(geo_data, City %in% input$city) })

  output$mapPlot <- renderPlot({
    d <- filtered()
    map("state", fill = TRUE, col = "gray92", border = "gray60")
    title(main = "City Distribution (Size = Population)")
    points(d$Longitude, d$Latitude, pch = 21, bg = adjustcolor(city_cols[d$City], 0.7),
      cex = d$Population / 150000)
    text(d$Longitude + 3, d$Latitude, labels = d$City, pos = 4, cex = 0.8, font = 2)
  }, width = 420, height = 300, res = 96)

  output$scatterPlot <- renderPlot({
    d <- filtered()
    plot(d$Avg_Temperature, d$Population / 1000, pch = 19, cex = 1.5, col = city_cols[d$City],
      xlim = c(65, 83), ylim = c(450, 750), xlab = "Average Temperature (F)",
      ylab = "Population (thousands)", main = "Temperature vs Population")
  })
}
```

```

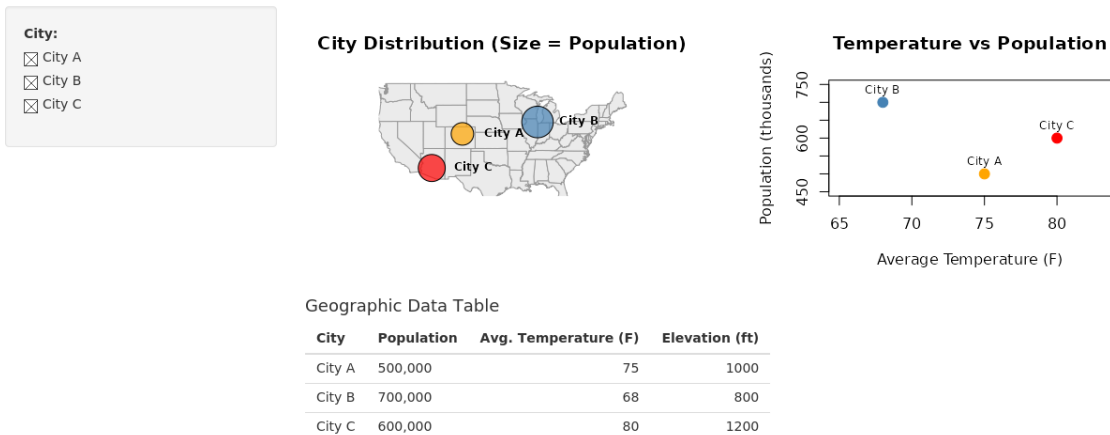
    text(d$Avg_Temperature, d$Population / 1000, labels = d$City, pos = 3, cex = 0.8)
  }, width = 420, height = 300, res = 96)

output$geoTable <- renderTable({
  d <- filtered()[, 1:4]
  d$Population <- formatC(d$Population, format = "d", big.mark = ",")
  colnames(d) <- c("City", "Population", "Avg. Temperature (F)", "Elevation (ft)")
  d
}, digits = 0)
}

shinyApp(ui = ui, server = server)

```

Geographic Data Dashboard



Output 4: Geographic Data Interactive Dashboard (R Shiny)

Observation: The dashboard shows the city map, the temperature vs population scatter plot, and the geographic data table together. Unticking a city in the City filter removes it from the map, the scatter plot, and the table at the same time.

Procedure

1. Create the geographic data frame with City, Population, Average Temperature, and Elevation.
2. Add assumed latitude and longitude values, draw the base map using `map("state")`, and plot the cities as bubbles with `points()` and `text()` labels.
3. Use `plot()` for the temperature vs population scatter plot, `abline()` with `lm()` for the trend line, and `cor()` for the correlation.
4. Format the population values, rename the columns with `colnames()`, and print the labelled table.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The geographic data was successfully visualized using a map chart, scatter plot, and table in R, and an interactive R Shiny dashboard was built combining all three views.

Key R Functions Used

Function	Purpose
<code>map()</code>	Draws the base geographic map (maps package).
<code>points()</code>	Plots the cities as bubbles on the map.
<code>text()</code> / <code>legend()</code>	Labels the cities and explains the bubbles.
<code>plot()</code>	Creates the scatter plot.
<code>abline()</code> / <code>lm()</code>	Adds a linear trend line.
<code>cor()</code>	Calculates the correlation coefficient.
<code>formatC()</code> / <code>colnames()</code>	Formats and labels the table.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>renderTable()</code>	Displays the filtered table on the dashboard.
<code>shinyApp()</code>	Runs the interactive dashboard.